

# Tiled Dirichlet-Process Gaussian Splatting for UAV Mapping with Calibrated Uncertainty

Aqi Dong

**Abstract**—Operational aerial-mapping decisions need trustworthy, spatially resolved uncertainty at survey scale. We present a fully Bayesian treatment of tiled large-scene splatting: independent Dirichlet-process Gaussian-mixture fits per tile are matched and merged exactly in closed form, then stitched by a partition-of-unity gate into one *normalized*, provenance-preserving scene-level predictive density, evaluated as a density, never as a renderer. On three Hessigheim 3D (H3D) LiDAR epochs (59–129 million points), that density matches the hard seam-deletion rules of current practice to within 0.02–0.03 nats per held-out point while deleting nothing. The normalized posterior is then spent: coverage at 68/90/95% is near-nominal on H3D (68% seam-band coverage 0.64–0.67); at a second site, all-points coverage replicates but seam coverage falls to 0.57. Predictive log-density ranks held-out error better than random (area under the sparsification error curve, AUSE, 0.16–0.20) but a model-free local-density ranker does better still, so we claim the uncertainty as a compact, calibrated, density-faithful surrogate, not as ranking power beyond sampling density; predictive confidence nonetheless stratifies mesh-referenced error monotonically. A null-calibrated model-derived level of detection holds its nominal level on stable ground (4.2% vs. nominal 5%; AUC 0.72) where a same-grid classical DEM-of-difference baseline flags 17–90% of the same cells, though the classical test remains the sharper pure ranker (AUC 0.82–0.97); the  $\approx 1$  m detection floor is a stated representation limit. In re-flight planning, posterior uncertainty matches a per-tile error oracle and beats a mass-only planner on comparable-mass tiles (+1.10 vs. +0.86 nats); where mass separates tiles, mass alone plans as well. The pipeline is CPU-only: a laptop fits 152 million points in 2,346 s.

**Index Terms**—Change detection, Dirichlet process, Gaussian splatting, LiDAR, UAV mapping, uncertainty quantification, variational inference.

## I. INTRODUCTION

Aerial mapping products increasingly feed operational decisions (infrastructure inspection, multi-epoch change monitoring, follow-up flight planning) that consume a reconstruction and then act on it. What they need is not photorealism but *trustworthy, spatially resolved uncertainty* at survey scale ( $10^8$  points per campaign) on hardware a field team actually has. Current practice supplies neither. Survey-scale scenes force partitioned processing, and the dominant large-scale splatting and neural-field pipelines resolve the resulting tile seams by geometric heuristics: hard cropping of out-of-tile primitives [1]–[3], unweighted consensus averaging [4], or hand-tuned image-space distance

blending [5]. As a result, no normalized global predictive density is defined, no contribution is weighted by posterior uncertainty, and duplicated boundary structure is deleted, never fused (Sec. II). Uncertainty, where it appears at all, is a post hoc overlay on a photometric optimizer [6], not a property of a generative model that can be calibrated and then spent.

This paper closes that gap for point-cloud splatting. Building on the DP-Splat model [7], a conjugate Dirichlet-process (DP) mixture over colored points fitted by closed-form coordinate ascent, we develop a fully Bayesian treatment of *tiled* large-scene fitting. In one sentence: partition-of-unity stitching with exact conjugate merging turns independent tile posteriors into one coherent, *normalized*, calibrated predictive density (at parity with hard crop/Voronoi deletion on predictive quality, but, unlike them, a well-defined probability model of the whole scene), and that density is what tells the operator where the reconstruction cannot be trusted, and where to fly next. Our contributions:

- 1) **Tiled conjugate inference with exact merging.** A partitioned DP-Splat pipeline: halo down-weighting, per-tile concentration bookkeeping ( $\sum_t \alpha_t = \alpha$ ), Hungarian matching of straddling components under a normal-inverse-Wishart (NIW) marginal-likelihood score with exact prior-subtracted merging, and partition-of-unity stitching (Sec. III). The matcher fires at production capacity, fusing, with exact provenance, the straddling structure that hard seam rules simply delete (Sec. V).
- 2) **A held-out, seam-aware evaluation, reported at parity.** To our knowledge, the first exact-conjugate-generative treatment of tiled large-scene splatting evaluated as a scene-level predictive density. On three Hessigheim 3D (H3D) epochs the stitched predictive matches the crop and Voronoi baselines to within 0.02–0.03 nats per held-out point (hard deletion is empirically near-costless here) while uniquely yielding a global density normalized up to a measured 0.020 tail deficit (mass 0.980 by numerical integration, Sec. IV-D). Credible intervals are calibrated at production capacity including within the seam bands on the H3D epochs; at the second site, all-points calibration replicates but seam-band calibration does not. The merge contributes exact model-order de-duplication and, by ablation, +0.14–0.35 nats of seam log-predictive across the three epochs (Sec. V).
- 3) **Uncertainty that is calibrated and then spent,** on three remote-sensing tasks, each with its controls measured (Sec. V). In error sparsification, the predictive negative

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log-density (NLPD) ranks held-out error better than random on real LiDAR but is outranked by a model-free local-density control, so we read the signal as a compact, calibrated, density-faithful surrogate, not as ranking power beyond sampling density. Null-calibrated per-cell change detection, with a capacity dose–response, holds the null that a same-grid classical DEM-of-difference (DoD) baseline badly violates. In re-flight planning, posterior uncertainty matches a per-tile error oracle and out-plans a mass-only control where per-tile recoverable mass is comparable; where it is not, mass dominates and the attribution is stated.

- 4) **Laptop-scale practicality with automatic capacity adaptation.** The pipeline runs on a laptop CPU at  $10^8$ -point scale (SMBU, 20 tiles: 152M points in 2,346 s at the production configuration; 169M in 863 s, 37.6 GB peak, at the lighter reference arm, whose smaller per-tile budget explains the shorter time on more points), and the inferred per-tile component count  $\hat{K}$  adapts capacity to data volume, deconfounded explicitly:  $\hat{K}$  tracks point count, not scene content (Secs. V and VI).

Throughout, claims are scaled to the evidence: the model is evaluated as a predictive density over colored points, never as a renderer, and where a signal carries a confound or a task hits an intrinsic floor, the paper quantifies the limitation rather than eliding it (Sec. VI).

## II. RELATED WORK

### A. Tiled Large-Scale Splatting and NeRF

Spatial partitioning is the accepted route to city-scale radiance-field reconstruction, and existing pipelines resolve the resulting seams geometrically: by hard spatial cropping of out-of-tile primitives (VastGaussian [1], Sec. 4.3; CityGaussian [2], Sec. 3.2 and its released merge script; Kerbl *et al.*'s hierarchy [3], whose Sec. 6.3 deletes a primitive trained by one chunk when it lies nearer another, a Voronoi cull), by ADMM consensus whose parameter average (Eq. 8b of DOGS [4]) weights position, quaternion, scale, and opacity with uniform  $1/K$  over a preassigned shared set, or by hand-tuned inverse-distance image blending (Block-NeRF [5], Sec. 4.3); Mega-NeRF [8], drawn on here for its partitioning scheme (and, in Sec. IV, as the source of the Mill 19 collection), answers each query by a single nearest-centroid submodule. In every case the merge rule is geometric and heuristic: none defines a normalized global predictive density, none weights contributions by posterior uncertainty, and duplicated boundary primitives are deleted, never fused. That is workable for rasterization, which carries no normalization constraint, but untenable for a generative model whose output must integrate to one. Two designs are direct foils for Sec. III: Block-NeRF's separate appearance-matching pass shows that models fit in isolation disagree at boundaries and must be reconciled, and DOGS's uniform  $1/K$  average ignores how much evidence each replica holds; our merge answers both at once, exactly, by adding prior-subtracted natural parameters on the pooled evidence.

### B. Bayesian Splatting

Jia *et al.* [6] attach NIW summaries to the standard photometric 3D Gaussian splatting (3DGS) optimizer [9]: pseudo-counts from an opacity–volume heuristic, the optimizer's running center as “observation,” the NIW scatter statistic held at zero; no assignments, no E-step, no joint evidence lower bound (ELBO), and the stick-breaking weights (Sec. III-A) acting purely as a pruning schedule: summaries the authors themselves call “optimization-driven approximations” (their Sec. 4). That concession motivates the opposite design, an exact conjugate generative treatment in which responsibilities, sufficient statistics, and a monotone ELBO are the inference itself, with Jia *et al.* as precedent for image-space calibration; ours is in point/density space. VBGS [10] is the closest inferential relative: a conjugate mixture on colored points fitted by coordinate-ascent variational inference (CAVI), but with a fixed finite component budget and no predictive use of its posterior (rendered, not queried as a density); its additive updates stream frames into one fixed-capacity global model, whereas we decompose space, so capacity adapts per tile and fits parallelize. VBGS-SLAM [11] propagates camera-pose uncertainty for indoor tracking, a different quantity from the map posterior used here. None of these systems addresses tiling, overlap mass, or model-order semantics. The DP-Splat model, its stick-breaking prior, and its CAVI and stochastic variational inference (SVI) are from our prior work [7]; this paper contributes the tiled inference, the exact merge, the stitched predictive, and the aerial evaluation.

### C. 3D Gaussian Splatting for Aerial Mapping

Partitioned aerial 3DGS pipelines (Yan *et al.* [12]; Ortho-3DGS [13]; ULSR-GS [14]; ARSGaussian [15]; Horizon-GS [16]) establish tiling as the accepted route to large scenes but handle boundaries with engineering heuristics (auxiliary supervision, view assignment) and leave component counts uninterpreted; benchmark studies (Atik [17]; Fadilah *et al.* [18], who find geometric accuracy uncorrelated with rendering quality, and whose “3DGS-MCMC” baseline is a densification sampler, not posterior inference) quantify error only post hoc against a reference cloud. None performs Bayesian inference, produces a predictive density, or attaches calibrated uncertainty to the map itself. TreeDGS [19] reweights primitives by an opacity-derived reliability score, a concrete downstream consumer of the per-component uncertainty our posterior provides in closed form; GU-GS [20] realizes “uncertainty” as a learned mask on the training loss, with no probabilistic semantics; and PRIMU [21] predicts per-Gaussian uncertainty with post hoc features and a per-scene learned regressor, setting the evaluation vocabulary (area under the sparsification error curve, AUSE; error–uncertainty correlation) we adopt. Neither of the latter two reports calibration, the property we evaluate directly via interval coverage.

### D. Uncertainty Evaluation, Distributed Nonparametric Inference, and Active View Selection

Our evaluation protocol assembles established instruments: sparsification analysis and AUSE from Ilg *et al.* [22], the

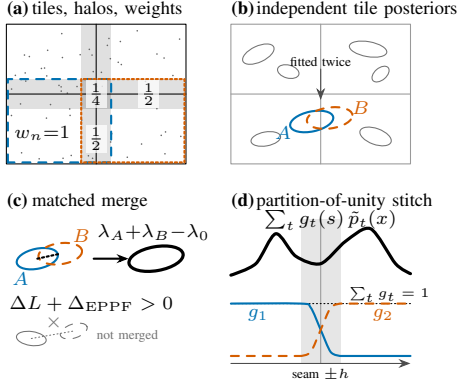


Fig. 1. Tiled DP-Splat pipeline. (a) Core tiles (solid grid) dilated by width- $h$  halos; a point in  $k_n$  dilated tiles gets weight  $w_n = 1/k_n$ , conserving evidence. (b) Independent weighted-CAVI fits under one shared NIW prior represent a straddling structure twice ( $A, B$ ). (c) Hungarian-matched pairs merge iff the closed-form Bayes factor plus DP partition term is positive ( $\Delta L + \Delta \text{EPPF} > 0$ ); accepted pairs combine exactly as  $\lambda_A + \lambda_B - \lambda_0$ ; unmatched components survive. (d) Raw bumps fall linearly across the halo band; the normalized gates  $g_t$  sum to one, blending tile sub-mixtures at global component weights into a single scene-level density.

area-under-random-gain metric (AURG) of Poggi *et al.* [23], and, for change detection, the M3C2 distance measure [24] with the error-propagated level of detection of M3C2-EP [25], which targets the pitfall (conflating surface roughness with positional uncertainty) that our posterior-mean covariance construction likewise avoids by design (Sec. IV-D); Huang and Qin [26] provide the plane-fit reference-based error convention our sparsification analysis adopts on H3D (Sec. IV-D). The merge machinery of Sec. III is likewise adapted, not invented: the closed-form merge-ratio score appears in memoized variational inference [27], partition-probability-thresholded component matching in streaming distributed Bayesian-nonparametric inference [28], Hungarian parameter matching in model aggregation [29], precision-weighted combination in consensus Monte Carlo [30], and the concentration bookkeeping  $\sum_t \alpha_t = \alpha$  that motivates interpreting the merged union against a single global DP in Williamson *et al.* [31] (Thm. 1); kernel stick-breaking [32] anchors the tiling: in the compact-kernel limit, spatially localized mixtures reduce to independent per-region DPs, the regime our tiles approximate. Finally, uncertainty-driven view selection for splatting (GAVIS [33], FisherRF [34], ActiveSplat [35], GauSS-MI [36]) is sequential, GPU-rasterized, photometric, and room-scale; our re-flight problem is batch, tile-level acquisition ranking from rasterization-free closed-form posterior scores, evaluated by held-out predictive density on a held-back image block’s triangulated points. GAVIS argues that parameter-space uncertainty underestimates unseen regions; a nonparametric predictive, whose prior reverts to a fat-tailed, low-capacity regime where data are absent, is a natural response, and we test it directly.

### III. METHOD

#### A. Base Model

DP-Splat [7] models a colored point cloud as a DP mixture of anisotropic Gaussians: point  $x_n = (s_n, c_n)$  has spatial coordinate  $s_n \in \mathbb{R}^{D_s}$  ( $D_s=3$ ) and color  $c_n \in \mathbb{R}^{D_c}$

( $D_c=3$ ), and conditional on its assignment  $z_n = k$  the modalities are independent Gaussians,  $s_n \sim \mathcal{N}(\mu_{k,s}, \Sigma_{k,s})$  and  $c_n \sim \mathcal{N}(\mu_{k,c}, \Sigma_{k,c})$ . Each modality  $m \in \{s, c\}$  carries a conjugate NIW prior  $\Sigma_{k,m} \sim \mathcal{IW}(\Psi_{0,m}, \nu_{0,m})$ ,  $\mu_{k,m} \mid \Sigma_{k,m} \sim \mathcal{N}(m_{0,m}, \Sigma_{k,m}/\kappa_{0,m})$ , and the mixture weights follow a truncated stick-breaking DP:  $v_k \sim \text{Beta}(1, \alpha)$  for  $k < T$ ,  $v_T := 1$ ,  $\pi_k = v_k \prod_{j < k} (1 - v_j)$ , with truncation level  $T$  the component budget. Inference is mean-field:  $q$  factorizes over assignments (responsibilities  $r_{nk}$ ), sticks (Beta), and per-component, per-modality NIW factors; CAVI cycles closed-form conjugate updates with a monotone ELBO, and natural-gradient SVI handles large  $N$ . Derivations are in [7]; we restate only what the tiled construction manipulates.

Two facts carry the rest of this section. First, the NIW is exponential family: writing  $\lambda(m, \kappa, \Psi, \nu) = (\kappa, \kappa m, \nu, \Psi + \kappa m m^\top)$  for its natural parameters, the variational posterior of component  $k$  in modality  $m$  is the prior plus that component’s expected sufficient statistics,

$$\lambda_{k,m} = \lambda_{0,m} + t_{k,m},$$

$$t_{k,m} = \left( N_k, \sum_n r_{nk} x_n^m, N_k, \sum_n r_{nk} x_n^m x_n^{m\top} \right), \quad (1)$$

with  $N_k = \sum_n r_{nk}$ . Second, the posterior predictive is a closed-form mixture of Student- $t$  products,

$$\hat{p}(x) = \sum_{k=1}^T \mathbb{E}_q[\pi_k] \prod_{m \in \{s, c\}} \text{St}(x^m; m_{k,m}, \tilde{\Psi}_{k,m}, \eta_{k,m}), \quad (2)$$

with  $\eta_{k,m} = \nu_{k,m} - D_m + 1$  and  $\tilde{\Psi}_{k,m} = \frac{\kappa_{k,m} + 1}{\kappa_{k,m} \eta_{k,m}} \Psi_{k,m}$ . Equation (2) is a normalized density; every evaluation in this paper is a functional of it or of the underlying posterior, never of a rendering.

#### B. Tiled Weighted Fitting

Aerial scenes of  $10^7$ – $10^8$  points do not fit in a single CAVI instance (the dense  $N \times T$  responsibility array alone is  $\sim 350$  GB at  $N=1.7 \times 10^8$ ,  $T=256$  in double precision), so we partition the ground plane: a  $G_x \times G_y$  grid of core rectangles exactly partitions the  $(x, y)$  bounding box (the vertical axis is never split), and each tile owns its core dilated by a halo of physical width  $h$ , so adjacent tiles share the halo points (Fig. 1a). Tiles are enumerated in a fixed Morton order (deterministic, locality-preserving indexing), and coordinates are centered in double precision so UTM-scale offsets never enter the natural-parameter algebra.

A point in  $k_n$  dilated tiles receives ownership weight  $w_n = 1/k_n$  in each, so  $\sum_t \sum_{n \in t} w_n = N$ : evidence is conserved, not duplicated. Each tile is fit by *weighted* CAVI: point  $n$ ’s likelihood and assignment factors are raised to the power  $w_n$ , counted  $w_n$  times with fractional counts allowed. The variational family is unchanged (the E-step remains the responsibility softmax); the weights enter only where the objective sums over points: the sufficient statistics

$$\begin{aligned} N_k &= \sum_n w_n r_{nk}, & \bar{x}_k &= \frac{1}{N_k} \sum_n w_n r_{nk} x_n, \\ S_k &= \sum_n w_n r_{nk} (x_n - \bar{x}_k)(x_n - \bar{x}_k)^\top, \end{aligned} \quad (3)$$

which feed the unmodified conjugate updates, and the point-sum ELBO terms, which each carry a factor  $w_n$  (the factor multiplies each point's entire ELBO contribution, entropy included, so it cancels in the assignment update; the softmax is untempered, exactly as the duplication identity requires). Two regression-tested identities pin the construction down:  $w_n \equiv 1$  reproduces the unweighted fit exactly, and integer  $w_n$  equals physically duplicating point  $n$  that many times.

The scene-level DP concentration  $\alpha$  (written  $\alpha_{\text{global}}$  below) is allocated across tiles in proportion to weighted evidence,

$$\alpha_t = \alpha_{\text{global}} \frac{1}{N} \sum_{n \in t} w_n, \quad \sum_t \alpha_t = \alpha_{\text{global}}, \quad (4)$$

a bookkeeping *motivated by* the gamma-process superposition theorem of Williamson *et al.* [31] (Thm. 1): Dirichlet-weighted superpositions of independent  $\text{DP}(\alpha_j, H)$  draws with  $\sum_j \alpha_j = \alpha$  are again  $\text{DP}(\alpha, H)$ . Three deviations place our construction outside its hypotheses: tile assignment is deterministic by location, not exchangeable; the sub-measures combine by deterministic gates (Sec. III-D), not Dirichlet weights; and the  $\alpha_t$  are set from realized counts. Equation (4) therefore guides the model-order accounting of Sec. III-E as an analogy, not a theorem-backed guarantee. All tiles share one global NIW prior  $\lambda_{0,m}$  per modality, built from the full-scene data scale by the base model's prior recipe: a precondition for the exact merge, whose algebra subtracts a single common prior.

Tile fits are embarrassingly parallel and never communicate. Tiles at or below an SVI threshold run full-batch weighted CAVI to a relative ELBO tolerance; larger tiles run weighted natural-gradient SVI (minibatch statistics scaled by  $w_n r_{nk}$ ) followed by *one exact full-batch weighted CAVI cycle*, so every final tile posterior equals prior plus full weighted statistics as in (1): the form the merge assumes, not SVI's stochastic blend of it. Configuration values are listed in Sec. IV. Full-CAVI tiles are audited for a monotone ELBO trace; SVI-path tiles only at the exact closing cycle (their subsampled trace is stochastic); outcomes in Sec. V-A.

### C. Exact Cross-Tile Matched Merge

Independent tile fits represent a surface that straddles a tile boundary twice: once per tile, each copy trained on that tile's down-weighted halo view. Rather than deleting one copy by a geometric rule, we test whether the two posteriors describe the same component and, if so, merge them exactly (Fig. 1b, c). Every ingredient is adapted from the distributed-Bayesian precedents of Sec. II-D; ours is the application to spatially tiled, fully independent fits with halo de-duplication and a single stitched global predictive.

*Candidates.* After discarding numerically empty components (weighted count  $N_k \leq n_{\min} = 1$ ), each pair of tiles with intersecting halo-dilated rectangles is scanned: a component pair  $(A, B)$  is a candidate if both posterior spatial means lie in the shared overlap box dilated by one further halo width. The gate is deliberately generous; the sign test below does the real rejection.

*Score.* The NIW log-normalizer is closed-form in the parameterization of (1):

$$A(\lambda) = \frac{D}{2} \log 2\pi - \frac{D}{2} \log \kappa + \frac{\nu D}{2} \log 2 + \log \Gamma_D\left(\frac{\nu}{2}\right) - \frac{\nu}{2} \log |\Psi|, \quad (5)$$

with  $\Gamma_D$  the multivariate gamma function. Because posterior naturals are prior plus statistics, the natural-space combination  $\lambda_A + \lambda_B - \lambda_0 = \lambda_0 + t_A + t_B$  is exactly the posterior a single component would have reached on the pooled statistics, and

$$\Delta L = A(\lambda_A + \lambda_B - \lambda_0) - A(\lambda_A) - A(\lambda_B) + A(\lambda_0), \quad (6)$$

summed over the two modalities, is the log Bayes factor of “one component generated both blocks” against “two independent components” (base-measure terms cancel: counts add). The partition side is the DP exchangeable partition probability function (EPPF) ratio of merged versus separate clusterings: since the DP EPPF is proportional to  $\alpha^{\#\text{clusters}} \prod_c \Gamma(N_c)$  [37], replacing clusters of sizes  $N_A, N_B$  by one of size  $N_A + N_B$  contributes

$$\Delta_{\text{EPPF}} = \log \Gamma(N_A + N_B) - \log \Gamma(N_A) - \log \Gamma(N_B) - \log \alpha_{\text{global}}, \quad (7)$$

the merge/no-merge partition term of Campbell *et al.* [28] (their Eqs. 13–14) specialized to the DP, evaluated at  $\alpha_{\text{global}}$  (never  $\alpha_t$ : the merged pair is an event of the *global* partition, whose concentration is  $\alpha_{\text{global}}$ ) and extended continuously to the soft counts by the gamma form.

*Decision and merge.* For each adjacent tile pair, a Hungarian assignment [38] maximizes  $\Delta L + \Delta_{\text{EPPF}}$  over the candidate mask; an assigned pair is accepted iff its total score is strictly positive. Connected components of the resulting cross-tile match graph form  $J$ -way merge sets, each combined in natural space,

$$\lambda_{\text{merged}} = \sum_{i=1}^J \lambda_i - (J-1) \lambda_0, \quad (8)$$

which counts the shared prior exactly once and equals the single-component posterior on the pooled weighted statistics for any  $J \geq 2$ : exact conjugate arithmetic, no moment matching. Unmatched components survive as distinct global atoms.

*Weights.* Stick-breaking Beta parameters are never added across tiles (labels and size-biased orderings differ per tile, the failure mode named by Campbell *et al.* [28]); the global sticks are rebuilt from the pooled soft counts in size-biased (descending) order:  $a_k = 1 + N_{(k)}$ ,  $b_k = \alpha_{\text{global}} + \sum_{j>k} N_{(j)}$ .

One sign in (7) deserves emphasis: beyond a few counts,  $\Delta_{\text{EPPF}}$  is *positive*, growing as  $\approx (N_A + N_B) H\left(\frac{N_A}{N_A + N_B}\right)$  ( $H$  the binary entropy); the DP's rich-get-richer preference for merging ( $\log \alpha_{\text{global}}$  subtracts only 4.6 at  $\alpha=100$ ), at its largest at the  $10^6$ – $10^7$ -point tiles used here. Rejection is carried by the likelihood term: when the two halo views' statistics disagree,  $\Delta L$  is negative at order  $N$  and empirically dominates the partition bonus, so the procedure defaults to *not* merging on discordant evidence and failure degrades toward the independent-tiles model, not spurious fusion; acceptance rates and their capacity dependence are reported in Sec. V. One caveat is in order: the decision is exact for the marginal-likelihood

comparison on the components' own (halo-local) statistics; we claim no global-ELBO monotonicity for the merged model, and we verify its quality only by held-out evaluation.

#### D. Partition-of-Unity Stitching

It remains to combine tile predictives into one scene-level density (Fig. 1d). Each tile  $t$  carries a separable bump over the ground plane,  $\text{raw}_t(s) = \prod_{d \in \{x,y\}} \rho_{t,d}(s_d)$ , where  $\rho_{t,d}$  equals 1 on the core interval, falls linearly to 0 across the width- $h$  halo band, and is 0 beyond. The gates are the normalized bumps

$$g_t(s) = \frac{\text{raw}_t(s)}{\sum_u \text{raw}_u(s)}, \quad (9)$$

which satisfy  $\sum_t g_t(s) = 1$ , a partition of unity everywhere in the scene bounding box by construction, including edges and corners (the denominator is  $\geq 1$ : every point lies in some core). The stitched predictive gates the per-tile *sub-mixtures*: writing  $\tilde{p}_t$  for (2) restricted to tile  $t$ 's components evaluated at their *global* post-merge mixture weights (Sec. III-C), un-renormalized within the tile,

$$\log p(x) = \log \sum_t g_t(s) \tilde{p}_t(x). \quad (10)$$

The global-weight convention is load-bearing: because the gates sum to one, a component carried by several tiles contributes  $\mathbb{E}_q[\pi_k] \sum_t g_t = \mathbb{E}_q[\pi_k]$  (counted once, never double), so (10), defined on the halo-dilated scene footprint and taken as zero outside it, has unit total mass up to Student- $t$  tail mass beyond the footprint; renormalizing  $\tilde{p}_t$  within tiles instead would inflate the mass to  $\approx \# \text{tiles}$  (Sec. IV-D). In the configuration where every tile carries one common predictive, (10) collapses to exactly that density. The acceptance tests verify the gate identity to  $10^{-12}$  on grids concentrated in the halo bands; verify unit total mass in that configuration by two independent numerical oracles (Gauss-Legendre quadrature and importance-sampled Monte Carlo); and, end to end, verify that two independently fitted tiles carrying their components at *global* mixture weights stitch to within 0.05 nats of a pooled single fit on held-out points. Beyond the pooled weight rebuild of Sec. III-C, stitching needs no inter-tile communication and ships regardless of merge decisions: the merge changes *what* the tiles agree on near seams, the gates govern *how* their densities combine.

#### E. $\hat{K}$ as Capacity Adaptation

Each tile reports its effective model order  $\hat{K}_t = \#\{k : N_k > n_{\min}\}$  on the weighted soft counts, recorded before merging; the global order after merging is  $\sum_t \hat{K}_t$  minus the duplicates removed by Sec. III-C, an accounting motivated by the concentration bookkeeping of (4) (with the caveats stated there). We interpret  $\hat{K}$  as *automatic per-tile capacity adaptation to data volume*: under a shared truncation budget  $T$ , tiles with little supported evidence occupy few components while heavily populated tiles recruit many, without per-tile tuning. We do *not* claim  $\hat{K}$  is a scene-content complexity map: on matched- $N$  controlled fits it shows no scene-content signal, and on real tiled scenes it tracks per-tile point count (Secs. V, VI-D).

## IV. EXPERIMENTAL SETUP

### A. Datasets

We evaluate on four public aerial datasets spanning native UAV LiDAR, UAV photogrammetry, and controlled synthetic photogrammetry. Figures show pipeline products, not dataset ground truth; we release code, configuration, and evaluation scripts that reproduce every number from the original data, but no derived products of the non-commercially licensed sources.

1) *Hessigheim 3D (H3D)*: The primary dataset [39]: UAV laser scanning (Riegl VUX-1LR, octocopter) over Hessigheim, Germany, at roughly 800 points/m<sup>2</sup>, with per-point RGB from oblique imagery and eleven manual semantic classes (CC BY-NC-SA 4.0; registration-gated). We use the three UAV epochs: March 2018 ( $59.4 \times 10^6$  training points plus a spatially disjoint  $14.5 \times 10^6$ -point validation area), November 2018 ( $128.9 \times 10^6$ ), and March 2019 ( $119.0 \times 10^6$ ). These support cross-epoch replication and multi-epoch change detection (Secs. V-A–V-E). H3D also ships a March 2018 reference mesh; its 43 labeled training-split tiles ( $9.0 \times 10^6$  faces after exact de-duplication of slab-boundary faces; 132,622 m<sup>2</sup>) serve as the absolute geometric reference (Sec. V-D). The mesh is the dataset authors' reconstruction from the same campaign's imagery and LiDAR: an independent processing chain over the same acquisition, not an independent survey.

2) *Mill 19 Rubble*: An ungated UAV image collection over a construction/debris site near Pittsburgh [8]: 1,678 4K images with shipped PixSfM-refined poses (publicly released; no formal license file, noted for completeness). No point cloud ships with it, so we build a sparse metric anchor ourselves, CPU-only (SIFT at  $4\times$  downscale, sequential-plus-spatial matching, COLMAP `point_triangler` [40] with poses held fixed, no bundle adjustment), yielding 1,750,606 colored points over a metrically coherent  $275 \times 318$  m site; this image-side scene carries the re-flight planning experiment (Sec. V-F).

3) *STPLS3D (synthetic subset)*: Synthetic aerial photogrammetry [41] (CC BY-NC-SA 3.0, form-delivered download): colored point clouds at  $\sim 0.1$  m spacing with 18 semantic classes, from simulated UAV flights over scenes ranging from empty terrain to dense urban. Content varies while acquisition is held constant, making it the natural controlled testbed for the capacity study: 29 scenes at matched  $N = 3 \times 10^6$  points across a concentration and prior-scale grid (261 controlled fits; Sec. VI-D).

4) *GauU-Scene V2 (SMBU)*: One scene from a DJI Matrice 300 / Zenmuse L1 RGB-LiDAR campaign [42] (access by e-mailed license request): the merged SMBU campus cloud of  $169.1 \times 10^6$  colored points, our largest single input, stressing the scaling claim and providing the second-site held-out check (Secs. V-A–V-B).

### B. Model Configuration

All experiments run the tiled DP-Splat pipeline of Sec. III on an Apple-silicon laptop (36 GB RAM, CPU only). Scenes are partitioned into a ground-plane grid targeting  $\sim 10^7$  points per tile (H3D:  $2 \times 3$  to  $2 \times 7$  tiles; SMBU: 20 tiles; Mill 19:  $10^5$ -scale tiles matched to its smaller anchor cloud), each tile

dilated by a 5 m halo with ownership down-weighting and concentration allocation as in Sec. III-B. Tiles above  $2 \times 10^6$  points are fitted by the stochastic variational schedule (batch size 65,536, three data epochs, step size  $\rho_t = (t+64)^{-0.7}$  as in [7]); smaller tiles run full CAVI to a relative ELBO tolerance of  $10^{-6}$ . (The SMBU production fit records SVI threshold  $5 \times 10^5$  and tile target  $8 \times 10^6$ . Its realized tiles are highly uneven, from near-empty to  $41.2 \times 10^6$  points: 15 ran the SVI schedule, the three smallest, each under  $4 \times 10^5$  points, ran full CAVI, and two near-empty tiles were skipped. The threshold difference matters only for the two tiles of  $1.2$  and  $1.6 \times 10^6$  points, which ran SVI where the H3D threshold would assign full CAVI.)

The *production configuration* for all headline experiments is  $\alpha_{\text{global}} = 100$ , truncation  $T = 256$  per tile, justified by the capacity study of Sec. V-B: relative to the low-capacity reference arm ( $\alpha_{\text{global}} = 1$ ,  $T = 64$ ), production improves held-out predictive quality, resolves the one calibration defect we observed, and activates the cross-tile matcher, while error ranking is insensitive to the choice. Both arms are reported wherever they differ materially; runtimes in Sec. V-A.

### C. Evaluation Protocols

1) *Held-out complement protocol*: In-sample density scores flatter any generative fit, so all headline predictive numbers use a hold-out: each H3D epoch (and, for the second-site check, SMBU) is refit on a recorded 90% subsample (fixed seed), and 200,000 evaluation points are drawn from the *exact complement* of the recorded fit indices under an independent seed: points the model has never seen, from the same flight. All methods are scored on the identical point set. The hold-out is sample-wise, not spatial: at  $\sim 800$  points/m<sup>2</sup> a complement point lies centimeters from training points, so absolute scores and coverage describe quasi-interpolative generalization within a flight; method *comparisons* are unaffected, and a spatially blocked protocol is an open item (Sec. VI-F). Each arm is likewise a single fixed-seed refit scored once, so Tables I and II carry no paired standard errors. Every metric is reported on all points and on the *seam subset*: points lying inside two or more halo-dilated tiles, i.e., within one halo width of a tile boundary (8.5% of the March 2018 hold-out;  $n = 16,941$ ).

2) *Seam baselines*: The stitched predictive is compared against the two hard-selection rules of current practice (Sec. II-A): *crop*, which truncates each tile's mixture to its owned box, and *Voronoi*, which assigns each component to its nearest tile center and deletes it elsewhere. Both are evaluated as normalized densities on the same per-tile posteriors, isolating the seam-handling rule itself: a controlled ablation on shared halo-down-weighted fits, not an end-to-end reproduction of any native pipeline, which would train each expanded cell at full weight before cropping; on the equal-area grids used here the two rules coincide exactly (verified; they differ on irregular partitions).

3) *Change-detection protocol*: The November 2018 and March 2019 production fits are compared on a shared 1 m ground grid (0.5 m and 2 m in ablation), retaining cells with at least 20 points in both epochs, under two complementary tests. (i) *Null calibration*: cells whose dominant class is impervious

in both epochs, with dominant-class purity above 0.9 in each ( $n = 17,592$ ), are physically stable ground; their exceedance of the nominal-5% threshold measures calibration directly. (ii) *Detection*: among cells with per-epoch purity above 0.6, cells whose dominant class changes between epochs (vegetation-excluded headline) serve as a labeled-change proxy, scored by ROC AUC and precision/recall at the threshold; the proxy undercounts true vertical change (same-class construction) and includes vertical-less transitions (vehicles), so precision at the threshold is not a false-alarm rate; the null test carries that role. Independently labeled epochs mean dominant-label flips can also arise from labeling inconsistency, boundary rasterization, or leaf-off/leaf-on appearance without physical change; the proxy is unvalidated and used only for ranking, never calibration. Of 113,020 valid cells, the 0.6 purity filter leaves 93,144 eligible; excluding cells whose dominant class in either epoch is Low Vegetation, Shrub, or Tree leaves 43,221 scored (734 positives). A registration term  $\sigma_{\text{reg}}$  enters the level-of-detection variance once (Sec. IV-D); results are reported at  $\sigma_{\text{reg}} = 0$  and, as a sensitivity, at 2 cm.

4) *Re-flight protocol (leakage-proofed)*: On Mill 19 Rubble the 1,678-image flight is split by acquisition order: the first 80% forms the survey, the trailing 20% block (336 images) the held-back re-flight pool, emulating an unfinished mission over contiguous new ground. Triangulated points are assigned to survey or pool by which image set observes them in at least two views (1,480,265 survey; 266,168 pool; ambiguous points discarded); an interleaved every-fifth-image split is the null control. After a robust bounding-box crop of the anchor (the per-axis 0.5–99.5 percentile box of the survey cloud, applied to both sets; it drops 42,785 survey and 14,148 pool outliers), a fixed 25% of the cropped pool (63,005 points) is reserved as the evaluation set *before* any granting, identically across arms, so no evaluation point enters any refit. Tiles are a ground-plane partition of the full cropped site's bounding box:  $2 \times 3$  in the 6-tile scenario,  $3 \times 4$  in the 12-tile scenario. Planner arms select the top quarter of tiles by acquisition score (nearest-integer rounding: 2 of 6 tiles, 3 of 12) and receive the remaining pool points there; controls are random tile subsets of the same size (5–8 seeded replicates) and a per-tile *oracle* ranked by the pool points' true held-out error contribution under the initial model; improvement is the change in mean held-out NLPD, baseline minus arm, on the fixed evaluation set.

### D. Metrics

1) *Held-out log-predictive*: Mean log-density per point (nats, i.e., natural-log units) of the six-dimensional (spatial  $\times$  color) stitched predictive on held-out points; higher is better; reported for all methods on both splits. Density comparisons require normalized densities, so the stitched mass is oracle-enforced: tile sub-mixtures carry their components' *global* weights, un-renormalized within tiles (a straddling component contributes  $\pi_k \sum_t g_t = \pi_k$ ), and importance-sampling integration of the production March 2018 model measures total mass 0.980 (SE 0.002; the deficit is Student- $t$  tail mass outside the dilated footprint). Renormalizing per tile instead inflates the mass to  $\approx \# \text{tiles}$  (5.62 measured), a

regression-guarded failure mode. The crop/Voronoi baselines renormalize to exactly unit mass, so the stitched rule carries a 0.020-nat ( $-\log 0.980$ ) tail-mass handicap into every overall comparison of Table I. Set against the production deficits there, the handicap roughly cancels the March 2018 gap ( $-0.017$ ) but not the other two ( $-0.031/-0.033$ ), which would shrink to roughly  $-0.011/-0.013$  if their unmeasured tail mass matches March 2018's; the reference-arm seam-band gaps are more than an order of magnitude larger and are not explained by it.

2) *Calibration coverage*: For each held-out point, responsibilities over all (tile, component) pairs are formed under the stitched predictive (conditioning on the full point, position and color), and the point is tested against each claiming component's central equal-tailed Student- $t$  interval at nominal 68/90/95% along each spatial axis; reported coverage is the responsibility-weighted mean of these indicators, averaged over axes, on both splits. This is a component-conditional calibration check, asking whether a point falls in the central interval of the components that claim it, not the mixture-marginal CDF; we make no theoretical seam-coverage claim.

3) *Sparsification*: *AUSE* and *AURG*: Following the sparsification-curve methodology [22], [23], 200,000 points are sampled *exactly* from the stitched spatial predictive (rejection sampling with the gates as acceptance probability), and each sample is scored by the reference-based error convention of Huang and Qin [26]:  $e$  is the orthogonal distance to the least-squares plane through the sample's six nearest neighbors in the full epoch cloud. Samples are removed by decreasing uncertainty  $u$  in 2% steps up to 98%, recording the mean  $|e|$  of the retained set, normalized by the full-set mean. *AUSE* (lower is better) is the area between the uncertainty-ordered and error-oracle curves; *AURG* (higher is better) that between the random-removal and uncertainty-ordered curves. Three  $u$  variants are reported: per-point NLPD under the stitched mixture (the headline ranker); the generating component's marginal predictive variances pooled by root mean square, written  $\sqrt{\text{Var}}$  (the location-queryable alternative); and, as a model-free control, the negative of the local point density  $6/(\frac{4}{3}\pi r_6^3)$ , with  $r_6$  the distance to the sample's sixth reference neighbor: the same neighborhood the error plane is fitted to. Density-stratified *AUSE* is reported alongside as the honest confound decomposition, and the *AUSE* machinery itself is verified in Sec. V-C. Unlike the predict-then-observe setting of [22], [23], the scored points are the model's own samples ( $u$  and  $e$  are functionals of the same point and model), so the ranking is model criticism, a quality-assurance ordering of reconstructed mass against the reference geometry, not a before-measurement error map (Sec. V-C).

4) *Geometric accuracy and completeness*: Against the March 2018 reference mesh (Sec. IV-A), both directions of the standard accuracy/completeness pair are scored in meters. *Accuracy* (model $\rightarrow$ mesh): 200,000 samples drawn from each seam rule's spatial predictive (stitched by the gate-rejection sampler above; crop/Voronoi ancestrally from their normalized mixtures), each scored by exact unsigned point-to-mesh distance (a certified nearest-triangle search, oracle-tested against brute force); the 1.2–1.5% of samples falling outside the mesh tiles' footprint are excluded. *Completeness* (mesh $\rightarrow$ model):

200,000 area-weighted mesh-surface samples, restricted to 1 m ground cells holding at least five points of the fit's own input cloud (70.3% kept; the mesh extends beyond the surveyed area), each scored by distance to the nearest of  $5 \times 10^5$  dense model samples, a discretization that overestimates distance to the model by about the dense-bank spacing (median 0.326 m), the resolution floor of the completeness fractions. Both directions are additionally stratified into terciles of the stitched model's own per-sample spatial predictive NLPD.

5) *Level of detection (LoD95)*: For change detection we adopt the significance-testing convention of the M3C2 lineage [24], [25]: a per-cell threshold

$$\text{LoD}_{95} = 1.96 \sqrt{\text{var}_z^A + \text{var}_z^B + \sigma_{\text{reg}}^2}, \quad (11)$$

where  $\text{var}_z^E$  is the vertical variance of the responsibility-weighted *posterior mean location* in epoch  $E$ , derived in closed form from component  $k$ 's spatial NIW posterior as  $\Sigma_{\mu,k} = \Psi_{k,s}/(\kappa_{k,s}(\nu_{k,s} - D_s - 1))$ . Using the mean-location covariance rather than the full predictive covariance avoids double-counting surface roughness in the threshold (the pitfall M3C2-EP was designed to fix [25]), and we verify the two covariances are distinct by Monte Carlo. Concretely, each cell is scored per epoch at its mean observed ground point  $q = (c_x, c_y, \bar{z}_E)$ : responsibilities  $w_{tk} \propto g_t(q) \mathbb{E}_q[\pi_k] \text{St}_3(q)$  over (tile, component) pairs (global post-merge weights, as in (10)) give  $\mu_z^E = \sum_{tk} w_{tk} m_{k,z}$ , with  $\text{var}_z^E$  by the law of total variance (weighted  $[\Sigma_{\mu,k}]_{zz}$  within components plus the spread of  $m_{k,z}$  between them);  $d = \mu_z^A - \mu_z^B$  is a vertical difference on a fixed grid (DoD-style, conflating horizontal displacement on slopes and facades with elevation change, unlike M3C2's normal-direction projection), and change is flagged where  $|d| > \text{LoD}_{95}$ . The *classical* baseline of Sec. V-E is the propagated-error DoD convention on identical cells, masks, and proxy sets:  $d = \bar{z}_A - \bar{z}_B$  from the raw per-cell points, and (11) with the empirical standard errors  $s_E^2/n_E$  of those means (within-cell sample variance over the  $n_E$  points) in place of the posterior mean-location variances; the arms differ only in the variance source.

## V. RESULTS

### A. Seam Handling: The Stitched Density at Parity with Hard Deletion

Table I reports the seam-rule comparison: held-out log-predictive per point, stitched versus crop and Voronoi, under the complement protocol of Sec. IV-C. At production capacity the rules are at *parity*: stitched trails crop by 0.02–0.03 nats/point overall, with seam-subset differences from  $+0.01$  on March 2018 to  $-0.09/-0.14$  on the other epochs. At the reference arm the overall gap is still only  $-0.11$ , but the seam bands show a real 0.56–0.66 nat stitched deficit (coarse components straddling seams), which production capacity closes.

The SMBU row is the second-site check: a production-capacity fit of a different sensor and site under the identical protocol. Overall, parity replicates ( $-0.05$  nats). At the seams it does not: the stitched seam deficit is  $-0.37$  nats, between H3D's production parity and the reference arm's

0.56–0.66-nat deficits. Production-capacity seam parity is therefore an H3D finding that the second site tempers, not a general property of the stitched rule (Sec. VI-A).

The mechanism deserves an honest reading. Hard rules *delete* straddling structure: at production capacity, cropping removes 20, 24, and 18 components from the merged models ( $\hat{K}$  841/1050/1108 to 821/1026/1090), and on the low-capacity validation-area fit the three deleted components include one carrying 261,000 points. Yet Table I shows the deletion is empirically near-costless here: after global renormalization, neighboring components absorb the deleted mass almost exactly. The case for the stitched partition of unity is therefore *not* accuracy: it is the only rule of the three that yields a provenance-preserving global density normalized up to a measured tail deficit (mass 0.980, oracle-measured, Sec. IV-D) without discarding fitted structure, and that density is what Sections V-B–V-F spend: calibrated intervals, the NLPD ranker, the model-derived LoD, and the yield acquisition score.

The cross-tile matcher fires only when models are rich enough to have straddling structure: at production capacity it accepts 42 pairs chaining into 37  $J$ -way merge sets on March 2018 ( $\hat{K}$  880→841), 29/31 pairs on the other epochs (1079→1050, 1139→1108), and 21 pairs chaining into 19 sets on the 20-tile SMBU production fit; the reference arm accepts zero on all full-epoch fits and SMBU, one on the validation area: the expected dominance of the likelihood term at  $10^7$ -point tiles (Sec. III-C), not a failure mode. Full-epoch fits enter only the runtime report below.

The merge’s predictive contribution is isolated by ablation: refitting each epoch with matching disabled reproduces the identical tile posteriors ( $\hat{K}$  880/1079/1139, the production runs’ pre-merge counts exactly) and drops the stitched seam log-predictive by 0.35/0.17/0.14 nats (Mar. 2018/Mar. 2019/Nov. 2018, the order of Table I, here and below), widening the stitched-vs-crop seam gap to  $-0.40/-0.33/-0.36$  nats where the merged models sit at  $+0.01/-0.09/-0.14$  (the gap widens by more than the stitched drop because the crop baseline, re-evaluated on the unmerged fits, itself gains 0.06–0.08 nats at seams): the exact merge, not the gating, closes most of the seam gap. Its overall contribution is  $+0.030/+0.012/+0.010$  nats, and coverage is essentially unchanged (68% all-points, March 2018: 0.696 merged, 0.699 without).

*a) Scale and runtime:* Everything above runs on the 36 GB laptop of Sec. IV-B: reference-arm fits of the full epochs (59.4, 119.0,  $128.9 \times 10^6$  points) complete in 274, 612, and 597 s; production fits of the recorded subsamples (53.5, 107,  $116 \times 10^6$ ) in 705, 1,344, and 1,878 s. The largest input, the 20-tile SMBU cloud, fits at production configuration on its recorded  $152 \times 10^6$ -point subsample in 2,346 s; the full  $169.1 \times 10^6$  points fit in 863 s at the reference configuration ( $T = 64$  against production’s 256: roughly a quarter of the per-tile work, hence the shorter time on more points), at a 37.6 GB peak resident set size, exceeding physical RAM via macOS memory compression. Held-out evaluation adds 12–40 s per H3D epoch (49.7 s for SMBU); the ELBO audits of Sec. III-B (monotone full-CAVI traces, one exact closing cycle per SVI tile) held on every tile.

## B. Calibration and the Capacity Study

Table II reports empirical central-interval coverage of the stitched predictive at nominal 68/90/95%. At production capacity coverage is near-nominal everywhere, *including* the seam bands; we report the small deviations rather than smooth them: 90% and 95% run mildly conservative, and 68% slightly under near seams. At SMBU, all-points coverage is near-nominal but the seam 68% falls to 0.570: the reference-arm-style seam undercoverage recurring at production capacity. Like seam parity, seam calibration at production capacity is an H3D result, not a demonstrated general one. Disaggregating the axis mean that Table II reports is revealing: at 68% the vertical axis runs conservative on all four production arms (0.74–0.79 all-points) while the horizontal axes sit at 0.64–0.66, and the seam deficits are carried almost entirely by the horizontal axes (0.48–0.52 on SMBU seams, against 0.71 vertical). The axis mean thus understates the directional structure of the seam miscalibration.

*a) Capacity study:* The reference arm exhibits one real calibration defect: 68% seam coverage of 0.496, 0.461, and 0.430, material undercoverage confined to the seam bands. Raising capacity from  $\hat{K} = 42$  to  $\hat{K} = 841$  on March 2018 resolves it (0.496→0.671), replicating on the other epochs (Table II; both arms are 90%-subsample refits under the identical protocol). The defect was a capacity artifact (coarse components straddling seams), not a flaw of the stitching; the same artifact drives the reference arm’s seam-band deficit against crop, which production capacity closes (Table I). The capacity increase also improves absolute held-out quality by +2.5 nats/point ( $-5.83$  versus  $-8.35$  on March 2018) and activates the cross-tile matcher, while leaving error ranking essentially unchanged;  $\hat{K}$  adapts to per-tile data volume automatically (disclaimers: Sec. VI-D).

## C. Sparsification: Uncertainty as an Error Ranker

First, the machinery cannot manufacture the result: the AUSE implementation is regression-tested to  $10^{-12}$  against an independent definition-level reference; everything below measures the model on misspecified real LiDAR.

On the real epochs the ranker is genuinely better than random on every one (Table II, Fig. 2; production AUSE 0.161–0.205, AURG 0.521–0.577). The ranking is capacity-insensitive (March 2018 AUSE 0.161 at  $\hat{K} = 841$  versus 0.165 at  $\hat{K} = 42$ ), so the residual oracle gap is not a resolution artifact.

The signal decomposes honestly. Predictive uncertainty correlates strongly with local point density ( $\rho(u, \text{density})$  of  $-0.74$ ,  $-0.70$ ,  $-0.67$  at production): sparser regions are genuinely reconstructed worse, which is causally meaningful but not to be mistaken for pure model structure. Density-tercile stratification localizes the residual: within-tercile AUSE is 0.28–0.50 at production, and the within-tercile gain over random concentrates in the *sparsest* tercile (AURG 0.19–0.29); the denser terciles sit essentially at random (AURG  $-0.01$  to  $0.03$ ), so the global ranking’s power rides substantially on the density component. The sparsest-tercile concentration holds at the reference arm as well, so we attribute the confound to the intrinsic mismatch between LiDAR point-density

TABLE I

HELD-OUT LOG-PREDICTIVE PER POINT (NATS; HIGHER IS BETTER) ON 200,000 COMPLEMENT POINTS PER FIT; ALL ROWS, BOTH ARMS, ARE THE 90%-SUBSAMPLE REFITS OF SEC. IV-C. STITCHED VS. THE CROP/VORONOI HARD BASELINES (IDENTICAL ON THESE EQUAL-AREA GRIDS);  $\Delta$  = STITCHED — CROP. SEAM COLUMNS: POINTS WITHIN ONE HALO WIDTH OF A TILE BOUNDARY (8.2–8.6% OF EACH HOLD-OUT). THE SMBU ROW IS THE SECOND-SITE PRODUCTION FIT (152M-POINT RECORDED SUBSAMPLE, 20 TILES; SEC. IV-A).  $\Delta$  IS COMPUTED BEFORE ROUNDING. SOURCES: \*\_EVAL.JSON.

| Config                     | Epoch/site | $\hat{K}$ | All points |           |          | Seam subset |           |          |
|----------------------------|------------|-----------|------------|-----------|----------|-------------|-----------|----------|
|                            |            |           | Stitched   | Crop/Vor. | $\Delta$ | Stitched    | Crop/Vor. | $\Delta$ |
| Production                 | Mar. 2018  | 841       | -5.834     | -5.818    | -0.017   | -5.921      | -5.931    | +0.010   |
|                            | Mar. 2019  | 1050      | -7.905     | -7.874    | -0.031   | -8.414      | -8.320    | -0.094   |
|                            | Nov. 2018  | 1108      | -7.358     | -7.325    | -0.033   | -8.016      | -7.881    | -0.135   |
|                            | SMBU       | 1165      | -7.909     | -7.857    | -0.052   | -8.899      | -8.532    | -0.367   |
| Reference ( $\alpha = 1$ ) | Mar. 2018  | 42        | -8.352     | -8.240    | -0.112   | -8.535      | -7.910    | -0.624   |
|                            | Mar. 2019  | 58        | -10.061    | -9.948    | -0.113   | -11.021     | -10.466   | -0.556   |
|                            | Nov. 2018  | 54        | -9.526     | -9.415    | -0.112   | -10.382     | -9.726    | -0.656   |

TABLE II

CALIBRATION AND ERROR RANKING (SOURCES AS TABLE I PLUS \*\_SPARSIFICATION.JSON). LEFT: EMPIRICAL COVERAGE OF NOMINAL 68/90/95% CENTRAL PREDICTIVE INTERVALS (AXIS-AVERAGED) ON THE HELD-OUT COMPLEMENT POINTS, ALL POINTS / SEAM SUBSET. RIGHT: SPARSIFICATION OF STITCHED-MODEL SAMPLES SCORED AGAINST THE FULL EPOCH CLOUD (SEC. IV-D): AUSE (LOWER BETTER), AURG (HIGHER BETTER), SPEARMAN CORRELATIONS WITH PER-POINT ERROR AND LOCAL DENSITY. SPARSIFICATION COLUMNS: H3D EPOCHS ONLY (THE REFERENCE-BASED ERROR CONVENTION OF SEC. IV-D IS DEFINED ON H3D).

| Config                     | Epoch     | Coverage (all/seam) |             |             | Sparsification |       |                |                         |
|----------------------------|-----------|---------------------|-------------|-------------|----------------|-------|----------------|-------------------------|
|                            |           | 68%                 | 90%         | 95%         | AUSE           | AURG  | $\rho(u,  e )$ | $\rho(u, \text{dens.})$ |
| Production                 | Mar. 2018 | 0.696/0.671         | 0.935/0.932 | 0.976/0.977 | 0.161          | 0.577 | 0.586          | -0.735                  |
|                            | Mar. 2019 | 0.684/0.643         | 0.930/0.907 | 0.972/0.964 | 0.177          | 0.557 | 0.580          | -0.700                  |
|                            | Nov. 2018 | 0.685/0.637         | 0.930/0.910 | 0.973/0.967 | 0.205          | 0.521 | 0.546          | -0.672                  |
|                            | SMBU      | 0.684/0.570         | 0.925/0.876 | 0.970/0.954 | —              | —     | —              | —                       |
| Reference ( $\alpha = 1$ ) | Mar. 2018 | 0.659/0.496         | 0.928/0.845 | 0.973/0.959 | 0.165          | 0.595 | 0.625          | -0.691                  |
|                            | Mar. 2019 | 0.644/0.461         | 0.928/0.817 | 0.975/0.951 | 0.213          | 0.532 | 0.575          | -0.640                  |
|                            | Nov. 2018 | 0.649/0.430         | 0.924/0.824 | 0.973/0.946 | 0.252          | 0.479 | 0.536          | -0.599                  |

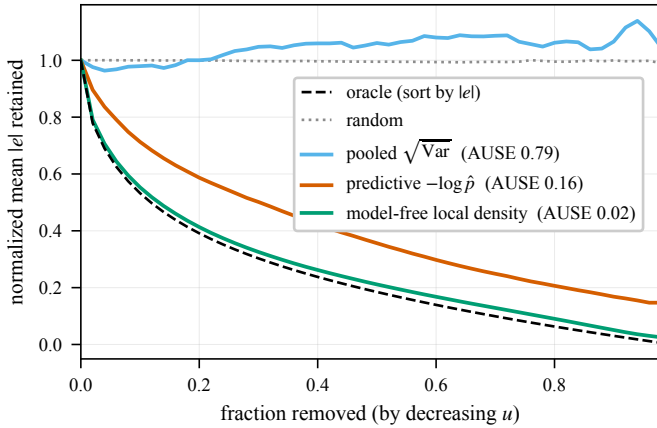


Fig. 2. Sparsification, March 2018, production configuration (pipeline product): 200,000 stitched-model samples scored against the full epoch cloud (Sec. IV-D); each curve removes by its own score (the oracle by  $|e|$ , random uniformly). Removing samples by decreasing predictive NLPD tracks the error oracle far better than random removal (AUSE 0.161, AURG 0.577); other epochs alike (Table II). The  $\sqrt{\text{Var}}$  curve is the location-queryable alternative, nearly uninformative as a ranker (AUSE 0.786, AURG -0.05); the model-free local-density control (AUSE 0.023, AURG 0.719) outranks the NLPD ranker and nearly attains the oracle (Sec. V-C).

heteroscedasticity and the Gaussian likelihood (Sec. VI-C). For completeness, the alternative per-point uncertainty  $\sqrt{\text{Var}}$  of Sec. IV-D is nearly uninformative (AURG -0.05 to +0.14 across the six runs), close to piecewise-constant within components. The distinction matters:  $\sqrt{\text{Var}}$  is the location-queryable uncertainty; the predictive NLPD is an a posteriori quality-assurance instrument for the reconstruction (Sec. IV-D), not a before-measurement error map.

The model-free control settles how much of that power is the model's own: ranking by raw local density of the reference cloud alone attains AUSE 0.023–0.024 and AURG 0.71–0.72 on the three epochs, decisively better than the NLPD ranker, and wins within every density tercile (AUSE 0.07–0.16 vs. 0.28–0.50); the two orderings agree at Spearman  $\rho = 0.67$ –0.74. The NLPD ranking is thus largely internalized sampling density, its non-density residual adding no within-tercile power. One caveat runs in the control's favor:  $e$  is the plane distance to the same six neighbors whose spacing defines the density, so sparse neighborhoods mechanically inflate  $e$  ( $\rho$  up to 0.92), flattering the control's near-oracle AUSE, though the within-tercile comparison, which coarsely controls this, still favors density. What the predictive retains is not ranking power but form: calibrated intervals and a  $\sim 10^3$ -component normalized density queryable without the  $10^7$ -point cloud (Sec. VI-C).

#### D. Absolute Geometric Accuracy Against the Reference Mesh

The sparsification ranker is scored against the same-epoch cloud; Table III answers the remaining absolute question: meters against the H3D March 2018 reference mesh, in both directions, under the protocol of Sec. IV-D. Globally the stitched model reconstructs the surface to mean 0.518 m (median 0.267, RMS 0.863, P90 1.32); completeness reaches 0.665 within 0.5 m and 0.951 within 1 m (the 0.25 m fraction, 0.254, sits at the 0.326 m sampling floor and is not interpretable below it). The choice of seam rule is immaterial at this scale: crop and Voronoi agree with the stitched model to within 3 mm in mean and 1.1 mm in median in both directions (crop's accuracy RMS runs 2.6 cm higher, a tail effect only), consistent with the parity of Table I.

TABLE III

GEOMETRIC ACCURACY AND COMPLETENESS (METERS) OF THE MARCH 2018 PRODUCTION FIT AGAINST THE H3D REFERENCE MESH, OVERALL AND STRATIFIED BY THE STITCHED MODEL'S OWN PREDICTIVE-NLPD TERCILES. ACCURACY: 197,508 MODEL SAMPLES, EXACT POINT-TO-MESH DISTANCE. COMPLETENESS: 140,574 DOMAIN-MASKED MESH-SURFACE SAMPLES TO THE NEAREST OF  $5 \times 10^5$  MODEL SAMPLES (RESOLUTION FLOOR 0.326 M; THE  $\leq 0.25$  M FRACTIONS ARE FLOOR-LIMITED). COMPLETENESS WITHIN 1 M IS 0.951 OVERALL (1.00/1.00/0.86 BY TERCILE). SOURCE: GEOACC\_MAR18.JSON.

|                                                           | Mean  | Med.  | RMS   | P90   | $\leq 0.25$ m | $\leq 0.5$ m |
|-----------------------------------------------------------|-------|-------|-------|-------|---------------|--------------|
| <i>Accuracy (model <math>\rightarrow</math> mesh)</i>     |       |       |       |       |               |              |
| All samples                                               | 0.518 | 0.267 | 0.863 | 1.320 | 0.479         | 0.690        |
| low NLPD                                                  | 0.144 | 0.108 | 0.215 | 0.289 | 0.856         | 0.978        |
| mid NLPD                                                  | 0.415 | 0.289 | 0.604 | 0.898 | 0.432         | 0.752        |
| high NLPD                                                 | 0.994 | 0.724 | 1.350 | 2.147 | 0.148         | 0.340        |
| <i>Completeness (mesh <math>\rightarrow</math> model)</i> |       |       |       |       |               |              |
| All samples                                               | 0.452 | 0.380 | 0.547 | 0.821 | 0.254         | 0.665        |
| low NLPD                                                  | 0.250 | 0.239 | 0.272 | 0.392 | 0.540         | 0.980        |
| mid NLPD                                                  | 0.402 | 0.387 | 0.435 | 0.624 | 0.182         | 0.739        |
| high NLPD                                                 | 0.704 | 0.655 | 0.796 | 1.092 | 0.041         | 0.276        |

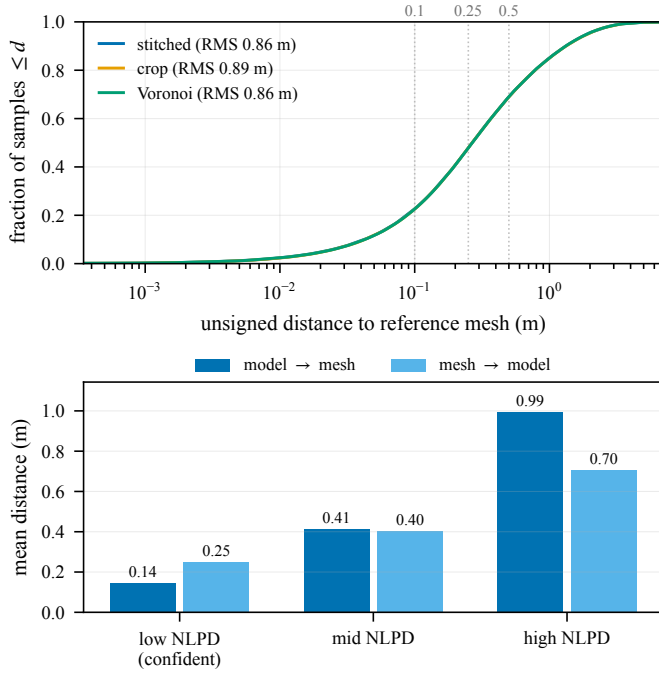


Fig. 3. Geometric accuracy against the H3D March 2018 reference mesh (pipeline product). Top: empirical CDF of unsigned model $\rightarrow$ mesh distance for the three seam rules (log axis; dotted lines at the 0.1/0.25/0.5 m thresholds); the rules are indistinguishable (curves overplot to line width). Bottom: mean distance by the stitched model's own predictive-NLPD tercile, both directions; the stratification is monotone, and the confident tercile is decimeter-grade.

The stratification is the operative finding (Table III, Fig. 3): ordering by the model's own predictive NLPD (no reference data in the loop) stratifies the error monotonically in both directions, with accuracy tercile means of 0.144/0.415/0.994 m. The confident third is decimeter-grade (85.6% of its samples within 0.25 m of the mesh, 97.8% within 0.5 m), and the model itself identifies it.

The honest scope: these are decimeter-to-meter numbers. Classical photogrammetric-LiDAR survey processing delivers centimeter-grade products on data of this quality (the reference mesh itself comes from such a chain), and the meter-band

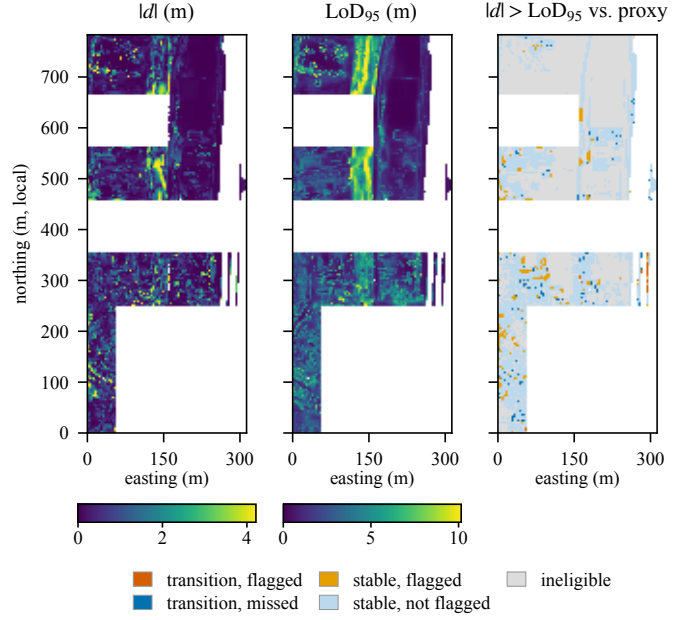


Fig. 4. Model-derived change detection, November 2018 vs. March 2019 (pipeline product): absolute inter-epoch elevation difference  $|d|$ , per-cell  $LoD_{95}$ , and flagged cells against the class-transition proxy on the 1 m grid; stable impervious cells calibrate the test (4.2% exceedance vs. nominal 5%). White cells (all panels) have fewer than 20 points in either epoch. Color scales are clipped at the 99th percentile; isolated flagged/transition cells are dilated by one cell for visibility.

high-NLPD tail is consistent with the  $\approx 1$  m level-of-detection floor of Sec. V-E (Sec. VI-B). What the stitched density offers is not centimeter accuracy but a calibrated map that knows where it is good: a self-identified decimeter-grade core, and meter-band tails the model itself flags.

#### E. Change Detection with a Model-Derived Level of Detection

The  $LoD_{95}$  test of Sec. IV-D applied to the November 2018/March 2019 production fits (Fig. 4) takes 21.3 s on the  $313 \times 783$ -cell grid (113,020 valid cells); on a synthetic 30 cm displacement slab it attains recall 1.000, null flag rate 0.000, and 0.06% displacement error, confirming the machinery before any real-data claim.

*Null calibration is delivered.* On the 17,592 physically stable impervious cells the exceedance rate is 4.2% against the nominal 5% (median  $|d|$  0.17 m; median  $LoD_{95}$  1.16 m), insensitive to the 2 cm registration term (exceedance rate and AUC both shift by  $\sim 10^{-4}$ ). *Detection is above chance:* on the vegetation-excluded transition proxy (734 positives, 42,487 negatives), AUC is 0.718 with recall 0.208 and precision 0.081 at the  $LoD_{95}$  threshold on 1 m cells; coarsening to 2 m cells raises AUC to 0.78 and recall to 0.35 with the null still calibrated (4.8%), while 0.5 m cells give AUC 0.68. Across valid cells median  $|d|$  is 0.28 m (95th pct. 2.48 m) vs. median  $LoD_{95}$  1.84 m; per the proxy caveat of Sec. IV-C, the false-alarm role belongs to the null test, not to precision at the threshold.

The dose-response (Table IV) settles whether more model is the answer. Tripling capacity sharpens the test: precision at the threshold rises from 0.081 to 0.207 and AUC from 0.718 to 0.776, at slightly lower recall (0.177 vs. 0.208) and a markedly

TABLE IV

CHANGE DETECTION ON 1 M CELLS: CAPACITY DOSE-RESPONSE OF THE MODEL-DERIVED TEST (BOTH ARMS AT  $\sigma_{\text{reg}} = 0$ ; HIGH-CAPACITY ARM:  $\alpha = 1000$ ,  $T = 1024$ ,  $\hat{K} = 3443/3525$  FOR MAR. 2019/NOV. 2018) BESIDE THE CLASSICAL PROPAGATED-ERROR DoD ON IDENTICAL CELLS, MASKS, AND PROXY (SEC. IV-D). SOURCES: `CHANGE{,_XRICH,_CLASSICAL}_NOV18_MAR19.JSON`.

|                                    | Model-derived |           | Classical DoD           |                                   |
|------------------------------------|---------------|-----------|-------------------------|-----------------------------------|
|                                    | Prod.         | High cap. | $\sigma_{\text{reg}}=0$ | $\sigma_{\text{reg}}=2\text{ cm}$ |
| Null exceedance (nom. 5%)          | 4.2%          | 0.9%      | 89.5%                   | 17.1%                             |
| AUC (veg-excluded)                 | 0.718         | 0.776     | 0.816                   | 0.973                             |
| Precision at LoD <sub>95</sub>     | 0.081         | 0.207     | 0.021                   | 0.050                             |
| Recall at LoD <sub>95</sub>        | 0.208         | 0.177     | 0.984                   | 0.971                             |
| Median null-cell LoD <sub>95</sub> | 1.16 m        | 1.04 m    | 1.3 mm                  | 3.9 cm                            |

more conservative null (0.9% against nominal 5%). But the LoD floor stays at  $\approx 1.0$  m: a representation limit (Sec. VI-B).

Table IV also delivers the promised classical yardstick, on literally identical cells. The propagated-error DoD posts a far sharper nominal threshold (median null-cell LoD<sub>95</sub> of 1.3 mm at  $\sigma_{\text{reg}} = 0$ , 3.9 cm at 2 cm, against 1.16 m) and, as a pure ranking score, separates the transition proxy better (AUC 0.816; 0.973 with the registration term). As a *detector* it is severely anti-conservative here: 89.5% (17.1% with the registration term) of the stable impervious cells exceed its nominal-5% threshold, against 4.2% for the model arm, because the per-cell standard error shrinks as  $1/\sqrt{n}$  (cells hold hundreds of points) while inter-epoch registration and roughness error do not. The model-derived LoD is thus deliberately conservative (recall 0.21 against 0.97, with a held null) and offers what the DoD cannot: the uncertainty rides in the fitted mixtures, so calibrated change maps need no per-cell access to both epochs' raw clouds at comparison time.

#### F. Variance-Guided Re-Flight Planning

On the Mill 19 Rubble anchor (Sec. IV-C;  $1.75 \times 10^6$  points, 1,678 images) the re-flight experiment asks whether the model's own posterior can plan the next flight; all arms are scored on one frozen evaluation set that no refit ever sees (Table V). The null control frames the block scenarios: with held-back imagery *interleaved* through the survey, no planner, including the error oracle, moved held-out NLPD by more than 0.04 nats at a 20% or 50% dose: re-flying covered ground buys nothing; the operational case is contiguous *new* ground.

*Act 1: posterior uncertainty finds where the model is wrong.* On the 6-tile site grid, with the unfinished 20% strip held back, the uncertainty planner selects exactly the tiles the error oracle selects and improves held-out NLPD by +1.10 nats, above every random grant ( $+0.45 \pm 0.17$ , 5 seeds; best  $+0.61$ ); the gain is local, not drift ( $+1.90$  inside the granted tiles). It also beats the mass-only control, which grants 8% more points (128,344 vs. 118,730) yet achieves only  $+0.86$ : the posterior finds *which* comparable-mass ground is badly modeled, not merely *where* data are recoverable.

*Act 2: raw uncertainty is mass-blind.* On the finer 12-tile grid, ranking tiles by raw mean NLPD still targets genuinely bad tiles ( $+1.88$  inside its grants) but selects tiles with almost no recoverable imagery: it grants only 19,734 pool points and moves the overall score by  $+0.14$ , below the random mean of  $+0.46 \pm 0.42$  (seed range  $+0.00$  to  $+0.99$ ).

TABLE V

RE-FLIGHT PLANNING ON MILL 19 RUBBLE (BLOCK HOLDBACK, NEW GROUND): IMPROVEMENT IN MEAN HELD-OUT NLPD (NATS; BASELINE — ARM), OVERALL AND WITHIN THE GRANTED TILES. RANDOM ARMS: MEAN  $\pm$  SAMPLE S.D. OVER SEEDED REPLICATES. <sup>†</sup>SELECTS THE SAME TILES AS EXPECTED YIELD, SO THE ROWS COINCIDE. <sup>‡</sup>SELECTS THE SAME TILES AS THE ERROR ORACLE (ROW OMITTED AS A DUPLICATE). SOURCES: `REFLIGHT_RUBBLE*_V2.JSON`, `REFLIGHT_RUBBLE_MASS{6,12}.JSON`.

| Scenario                      | Planner                  | $\Delta$ overall | $\Delta$ granted |
|-------------------------------|--------------------------|------------------|------------------|
| Block, 6 tiles<br>(2 grants)  | uncertainty <sup>‡</sup> | +1.10            | +1.90            |
|                               | mass only                | +0.86            | +1.37            |
|                               | random (5 seeds)         | $+0.45 \pm 0.17$ | $+1.45 \pm 0.54$ |
| Block, 12 tiles<br>(3 grants) | raw NLPD ranking         | +0.14            | +1.88            |
|                               | expected yield           | +1.15            | +1.45            |
|                               | mass only <sup>†</sup>   | +1.15            | +1.45            |
|                               | error oracle             | +0.55            | +1.96            |
|                               | random (8 seeds)         | $+0.46 \pm 0.42$ | $+1.15 \pm 0.52$ |

*Act 3: where mass separates tiles, mass dominates.* Scoring tiles by *expected yield* (mean tile NLPD  $\times$  grantable pool mass; the NLPD factor is available before flying, while the mass factor, measured here from the held-back block, stands in for the coverage-based yield estimate a planner would form in advance) resolves the failure:  $+1.15$  nats held-out (Table V), above all eight random draws (best  $+0.99$ ), and raises the fraction of evaluation points within 0.5 m of a model sample by  $+0.09$ . But so does recoverable mass alone: the mass-only control selects the identical tiles and matches  $+1.15$  exactly. On this grid the posterior factor adds no selection signal, and the margin over the error oracle (itself at  $+0.55$ ) is carried by the mass term: the oracle's grants carry 62,387 pool points against the yield planner's 157,739. Read together with Act 1: mass is the first-order planning signal, and the posterior's measured value concentrates exactly where mass stops discriminating.

## VI. DISCUSSION AND LIMITATIONS

### A. What the Stitched Density Does and Does Not Buy

Table I is deliberately a parity result. Stitching does *not* buy nats: a practitioner needing only a rasterized product loses nothing by cropping. It buys the object itself, a normalized global density with every fitted component's evidence intact and duplication resolved by an explicit exact merge, not implicit deletion. The crop baseline is moreover charitable (globally renormalized, where deployed pipelines skip it), and deletion's near-zero cost is an empirical fact of these scenes, not a guarantee: it grows in principle with the mass straddling seams, a regime this study does not test. The second site cuts the other way too: SMBU's production seam bands show a  $-0.37$ -nat stitched deficit and 68% seam coverage of 0.570 (Secs. V-A–V-B), so neither seam parity nor seam calibration of the stitched rule travels automatically across sites; both must be checked per campaign.

### B. The Level-of-Detection Floor Is Intrinsic

The change-detection level of detection saturates at approximately 1.0 m and no configuration we tried lowers it: the floor survives the  $\sim 3\times$  capacity increase of Table IV, cell sizes 0.5–2 m, and a 2 cm registration term, even as ranking and precision improve. We conclude it is a representation

limit of Gaussian mixtures on heterogeneous urban vertical structure: roof edges, facades, and vegetation within one component's support inflate the posterior mean-location variance the LoD is built from. We expect no tuning to remove it; sub-meter sensitivity would require a likelihood modeling vertical layering explicitly. The floor is consistent with the measured absolute accuracy: model-to-mesh error is decimeter-grade only in the confident tercile and meter-band in the worst (Sec. V-D), an order of magnitude above the centimeter grade classical survey processing achieves on data of this quality. For the LoD itself the classical yardstick is now measured (Sec. V-E): the same-grid propagated-error DoD posts nominal mm–cm thresholds but flags 17–90% of stable ground, so the  $\approx 1$  m floor is the price of a held null level here, not mere bluntness; whether a full M3C2-EP error budget with explicitly modeled alignment uncertainty [25] can hold the null at centimeter LoDs is the open classical arm (Sec. VI-F).

### C. The Density Component of the Uncertainty Signal

As quantified in Sec. V-C, the predictive-NLPD error ranking is largely internalized sampling density (a model-free density ranker outranks it, with rank agreement  $\rho = 0.67$ – $0.74$ ), and the within-tercile gain over random concentrates in the sparsest tercile; the pattern's replication across a  $20\times$  capacity change makes this intrinsic to the mismatch between LiDAR sampling heteroscedasticity and the Gaussian observation model, not a resolution artifact. An operator holding the raw cloud should rank by density; what the predictive adds is the calibrated, compact form (Sec. V-C), and its ranking should be read as “sparse and poorly explained,” not as a density-free model-error signal.

### D. What $\hat{K}$ Is and Is Not

We report the deconfounding of  $\hat{K}$  as a finding. On the STPLS3D matched- $N$  grid (29 scenes  $\times \alpha \in \{0.1, 1, 10\} \times 2$  seeds, all at  $N = 3 \times 10^6$  points), realized complexity at  $\alpha = 1$  falls in the narrow band  $\hat{K} \in [18, 25]$  from empty terrain to dense urban, with no content signal:  $\rho(\hat{K}, \text{class-entropy}) = 0.13$  ( $p = 0.34$ ),  $\rho(\hat{K}, \text{surface variation}) = -0.04$ ,  $\rho(\hat{K}, \text{density}) = 0.00$ ; the  $\alpha = 10$  arm is truncation-pinned at  $T = 64$ , and a prior length-scale sweep (29 scenes  $\times$  three scale factors,  $0.03$ – $0.3\times$  the default, at  $\alpha = 1$  and a single seed; the main grid's default-scale arm supplies the  $1\times$  endpoint) leaves per-scene  $\hat{K}$  unchanged to within one component (5 of 29 scenes shift by exactly  $\pm 1$ ). In total 261 controlled fits, the 174-fit matched- $N$  grid plus the 87-fit sweep: under truncated stick-breaking CAVI at matched  $N$ , realized complexity is set by component-death dynamics and data volume, consistent with the base model's asymptotics [7]; on the H3D tiles,  $\rho(\hat{K}, \text{tile point count}) = 0.97/0.82/0.85$  (full-epoch reference fits; production refits  $0.77$ – $0.98$ ). We claim  $\hat{K}$  only as automatic, data-volume-adaptive capacity control (useful, since a fixed budget as in VBGS [10] over-provisions empty tiles and starves dense ones, Fig. 5) and disclaim any reading of  $\hat{K}$  as a map of scene content.

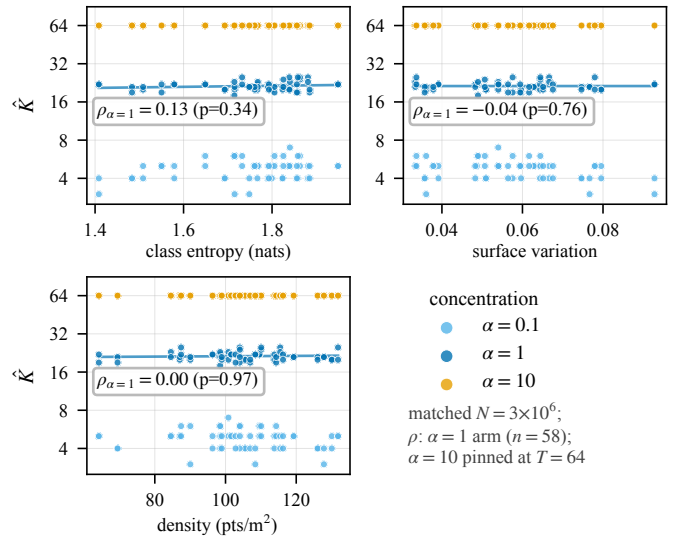


Fig. 5.  $\hat{K}$  against scene-content descriptors on the STPLS3D matched- $N$  grid (pipeline product; 29 scenes  $\times$  2 seeds per concentration arm, annotated  $\rho$  and least-squares trend lines from the  $\alpha=1$  arm; surface variation is the local-PCA smallest-eigenvalue ratio  $\lambda_{\min}/\sum_i \lambda_i$  over  $k=16$  nearest neighbors in a fixed  $5 \times 10^4$ -point uniform subsample per scene); the association is null, while the  $\alpha=10$  arm is truncation-pinned at  $T=64$ ; on real tiled scenes  $\hat{K}$  tracks per-tile point count ( $\rho$  up to 0.98, Sec. VI-D).

### E. Scope of the Re-Flight Study

The Mill 19 planning results rest on a sparse COLMAP anchor [8]: a planning-scale demonstration on held-back imagery, with no flights executed and no absolute-scale claims beyond the anchor's metric coherence. The mass-only control (grant by pool mass, ignoring the model) has been run in both scenarios (Sec. V-F): the posterior factor is dispensable for mass-separable grants and demonstrably valuable for mass-comparable ones; whether that margin persists on dense photogrammetric products is left to future work (Sec. VII).

### F. Open Protocol Questions

Cross-tile merge acceptance is rare at  $10^7$ -point tiles under  $\alpha = 1$  (Sec. V-A), consistent with dominance of the likelihood term in the sign test at large  $N$  (Sec. III-C); a halo-width and acceptance-threshold ablation with an ELBO-based criterion is left to future work, as are hierarchical Dirichlet-process (HDP) sharing across tiles, a spatially blocked hold-out protocol (Sec. IV-C), a full M3C2-EP error budget on the same cells (the plain propagated-error DoD arm is measured in Sec. V-E), and external validation of the class-transition change proxy. Finally, we claim no theoretical seam-coverage guarantee: calibration is an empirical, per-campaign check (Secs. V-B, VI-A).

## VII. CONCLUSION

Partition-of-unity stitching with exact conjugate merging turns independent per-tile posteriors into a single normalized, provenance-preserving predictive density that matches crop/Voronoi seam deletion on held-out quality (within  $0.02$ – $0.03$  nats/point on three H3D epochs) while deleting nothing, with near-nominal coverage including at seams; at the second

site, overall parity and all-points calibration replicate while the seam bands fall short, and we say so. That normalized density, not a nat advantage, is the contribution: it makes the calibrated intervals, NLPD error ranking, model-derived level of detection, and acquisition score well-defined, and it pays where spent: predictive confidence stratifies mesh-referenced accuracy monotonically, isolating a self-identified decimeter-grade core (0.14 m mean vs. 0.52 m overall); the model-derived LoD holds its null where a same-grid classical DoD flags 17–90% of stable ground (though the DoD remains the sharper pure ranker); posterior-guided re-flight matches a per-tile error oracle and beats a mass-only control on comparable-mass grants (+1.10 vs. +0.86 nats), with the mass-dominated regime attributed, and the exact merge is what holds the stitched density at seam parity (+0.14–0.35 nats at seams in ablation). All of this runs on a laptop at  $10^8$ -point scale. Future work: halo-width and merge-threshold ablations, a full M3C2-EP error budget for the change test, HDP cross-tile sharing, and a GPU dense phase extending to photogrammetric sources with LiDAR-referenced validation on GauU-Scene.

#### DATA AND CODE AVAILABILITY

All four datasets are publicly accessible from their originators [8], [39], [41], [42] under the access and license terms noted per dataset in Sec. IV-A. In keeping with the non-commercial license terms, no derived point clouds or fitted models are redistributed; the complete pipeline, configuration, evaluation scripts, and aggregate result records reproducing every number and figure from the original data are available at <https://github.com/archiedong/dp-splat-uav>.

#### REFERENCES

- [1] J. Lin *et al.*, “VastGaussian: Vast 3D Gaussians for large scene reconstruction,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, arXiv:2402.17427.
- [2] Y. Liu, H. Guan, C. Luo, L. Fan, J. Peng, and Z. Zhang, “CityGaussian: Real-time high-quality large-scale scene rendering with Gaussians,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Springer, 2024, arXiv:2404.01133.
- [3] B. Kerbl, A. Meuleman, G. Kopanas, M. Wimmer, A. Lanvin, and G. Drettakis, “A hierarchical 3D Gaussian representation for real-time rendering of very large datasets,” *ACM Trans. Graph.*, vol. 43, no. 4, 2024, arXiv:2406.12080.
- [4] Y. Chen and G. H. Lee, “DOGS: Distributed-oriented Gaussian splatting for large-scale 3D reconstruction via Gaussian consensus,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, 2024, arXiv:2405.13943.
- [5] M. Tancik *et al.*, “Block-NeRF: Scalable large scene neural view synthesis,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, arXiv:2202.05263.
- [6] G. Jia, V. Appia, J. Huang, and X. Wang, “Rendering-aware Bayesian 3D Gaussian splatting with native uncertainty and adaptive complexity control,” *arXiv preprint arXiv:2607.05522*, 2026.
- [7] A. Dong, “DP-Splat: Bayesian nonparametric complexity control for Gaussian splatting,” *arXiv preprint arXiv:2607.10912*, 2026.
- [8] H. Turki, D. Ramanan, and M. Satyanarayanan, “Mega-NeRF: Scalable construction of large-scale NeRFs for virtual fly-throughs,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, arXiv:2112.10703.
- [9] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3D Gaussian splatting for real-time radiance field rendering,” *ACM Trans. Graph.*, vol. 42, no. 4, pp. 1–14, Jul. 2023.
- [10] T. Van de Maele, O. Çatal, A. Tschantz, C. L. Buckley, and T. Verbelen, “Variational Bayes Gaussian splatting,” in *Proc. NeurIPS Workshop on Bayesian Decision-Making and Uncertainty*, 2024, arXiv:2410.03592.
- [11] Y. Zhu, Y. Zhang, J. Xu, and W. Ren, “VBGS-SLAM: Variational Bayesian Gaussian splatting simultaneous localization and mapping,” *arXiv preprint arXiv:2604.02696*, 2026.
- [12] K. Yan, H. Wang, Z. Li, Y. Wang, S. Li, and H. Yang, “A large-scale 3D Gaussian reconstruction method for optimized adaptive density control in training resource scheduling,” *Remote Sens.*, vol. 17, no. 23, p. 3868, 2025.
- [13] J. Yang, Z. Cai, T. Wang, T. Ye, H. Gao, and H. Huang, “Ortho-3DGS: True digital orthophoto generation from unmanned aerial vehicle imagery using the depth-regulated 3D Gaussian splatting,” *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 10972–10994, 2025.
- [14] Z. Li *et al.*, “ULSR-GS: Urban large-scale surface reconstruction Gaussian splatting with multi-view geometric consistency,” *ISPRS J. Photogramm. Remote Sens.*, vol. 230, pp. 861–880, 2025.
- [15] Y. Yao *et al.*, “ARSGaussian: 3D Gaussian splatting with LiDAR for aerial remote sensing novel view synthesis,” *ISPRS J. Photogramm. Remote Sens.*, vol. 231, pp. 288–306, 2026.
- [16] L. Jiang *et al.*, “Horizon-GS: Unified 3D Gaussian splatting for large-scale aerial-to-ground scenes,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025.
- [17] M. E. Atik, “Comparative assessment of neural radiance fields and 3D Gaussian splatting for point cloud generation from UAV imagery,” *Sensors*, vol. 25, no. 10, p. 2995, 2025.
- [18] W. A. Fadilah, A. Murtiyoso, T. Landes, and P. Grussenmeyer, “Metric assessment of 3D Gaussian splatting for UAV-based urban heritage reconstruction,” in *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. XLVIII-2/W12-2026, 2026, pp. 143–150.
- [19] B. Shaheen *et al.*, “TreeDGS: Aerial Gaussian splatting for distant DBH measurement,” *Remote Sens.*, vol. 18, no. 6, p. 867, 2026.
- [20] B. Ding, J. Liu, X. Yin, Z. Yang, Y. Luo, and D. Zhang, “GU-GS: Gaussian splatting-based geometry refinement and uncertainty-aware learning method for DSM generation from satellite imagery,” *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 1–15, 2026.
- [21] T. Gottwald, E. Heinert, P. Stehr, C. J. Galappaththige, and M. Rottmann, “PRIMU: Uncertainty estimation for novel views in Gaussian splatting from primitive-based representations of error and coverage,” *arXiv preprint arXiv:2508.02443*, 2025.
- [22] E. Ilg *et al.*, “Uncertainty estimates and multi-hypotheses networks for optical flow,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Springer, 2018.
- [23] M. Poggi, F. Aleotti, F. Tosi, and S. Mattoccia, “On the uncertainty of self-supervised monocular depth estimation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020.
- [24] D. Lague, N. Brodu, and J. Leroux, “Accurate 3D comparison of complex topography with terrestrial laser scanner: Application to the Rangitikei canyon (N-Z),” *ISPRS J. Photogramm. Remote Sens.*, vol. 82, pp. 10–26, 2013.
- [25] L. Winiwarter, K. Anders, and B. Höfle, “M3C2-EP: Pushing the limits of 3D topographic point cloud change detection by error propagation,” *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 240–258, 2021.
- [26] D. Huang and R. Qin, “Uncertainty estimation for photogrammetric point clouds of UAV imagery,” in *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. XLVIII-G-2025, 2025, pp. 657–664.
- [27] M. C. Hughes and E. B. Sudderth, “Memoized online variational inference for Dirichlet process mixture models,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 26, 2013.
- [28] T. Campbell, J. Straub, J. W. Fisher III, and J. P. How, “Streaming, distributed variational inference for Bayesian nonparametrics,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 28, 2015, arXiv:1510.09161.
- [29] M. Yurochkin, M. Agarwal, S. Ghosh, K. Greenewald, and T. N. Hoang, “Statistical model aggregation via parameter matching,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019, arXiv:1911.00218.
- [30] S. L. Scott, A. W. Blocker, F. V. Bonassi, H. A. Chipman, E. I. George, and R. E. McCulloch, “Bayes and big data: The consensus Monte Carlo algorithm,” *Int. J. Manage. Sci. Eng. Manage.*, vol. 11, no. 2, pp. 78–88, 2016.
- [31] S. Williamson, A. Dubey, and E. P. Xing, “Parallel Markov chain Monte Carlo for nonparametric mixture models,” in *Proc. 30th Int. Conf. Mach. Learn. (ICML)*, ser. Proc. Mach. Learn. Res., vol. 28, no. 1, 2013.
- [32] D. B. Dunson and J.-H. Park, “Kernel stick-breaking processes,” *Biometrika*, vol. 95, no. 2, pp. 307–323, 2008.
- [33] S. Xue, J. Dill, D. Ahuja, P. Dellaert, P. Tsotras, and D. Xu, “Uncertainty-driven 3D Gaussian splatting active mapping via anisotropic visibility field,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, arXiv:2605.30342.

- [34] W. Jiang, B. Lei, and K. Daniilidis, "FisherRF: Active view selection and mapping with radiance fields using Fisher information," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, ser. Lect. Notes Comput. Sci. Springer, 2024, pp. 422–440.
- [35] Y. Li *et al.*, "ActiveSplat: High-fidelity scene reconstruction through active Gaussian splatting," *IEEE Robot. Autom. Lett.*, vol. 10, no. 8, pp. 8099–8106, 2025.
- [36] Y. Xie, Y. Cai, Y. Zhang, L. Yang, and J. Pan, "GauSS-MI: Gaussian splatting Shannon mutual information for active 3D reconstruction," *arXiv preprint arXiv:2504.21067*, 2025.
- [37] C. E. Antoniak, "Mixtures of Dirichlet processes with applications to Bayesian nonparametric problems," *Ann. Statist.*, vol. 2, no. 6, pp. 1152–1174, 1974.
- [38] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Res. Logist. Quart.*, vol. 2, no. 1–2, pp. 83–97, Mar. 1955.
- [39] M. Kölle *et al.*, "The Hessigheim 3D (H3D) benchmark on semantic segmentation of high-resolution 3D point clouds and textured meshes from UAV LiDAR and multi-view-stereo," *ISPRS Open J. Photogramm. Remote Sens.*, vol. 1, p. 100001, 2021.
- [40] J. L. Schönberger and J.-M. Frahm, "Structure-from-motion revisited," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016.
- [41] M. Chen *et al.*, "STPLS3D: A large-scale synthetic and aerial photogrammetry 3D point cloud dataset," in *Proc. 33rd Brit. Mach. Vis. Conf. (BMVC)*, 2022.
- [42] B. Xiong, N. Zheng, J. Liu, and Z. Li, "GauU-Scene V2: Assessing the reliability of image-based metrics with expansive Lidar image dataset using 3DGS and NeRF," *arXiv preprint arXiv:2404.04880*, 2024.

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